

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Project Report

SQL • Python • Power BI

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1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across multiple product categories. The primary goal is to uncover insights into spending patterns, customer segments, product preferences, discount behaviour, shipping choices, and subscription activity so that the findings can support practical business decisions.

The workflow combines Python-based data preparation and exploratory analysis, SQL-based business analysis, and an interactive Power BI dashboard for visual communication.

2. Dataset Summary

Attribute	Description
Rows	3,900
Columns	18
Customer demographics	Age, Gender, Location, Subscription Status
Purchase details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
Missing data	37 values in the Review Rating column

The dataset contains customer-level transactional records. The analysis uses both numeric fields (such as age, purchase amount and review rating) and categorical fields such as category, gender, subscription status and shipping type.

3. Exploratory Data Analysis using Python

The initial stage focused on understanding the dataset's structure and quality before loading the cleaned data into the database for SQL analysis.

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to inspect the structure and `df.describe()` for summary statistics.
- Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Column Standardisation: Renamed columns to snake_case for better readability and documentation.
- Feature Engineering: Created an `age_group` column by binning customer ages.
- Feature Engineering: Created `purchase_frequency_days` from the purchase-frequency information.
- Data Consistency Check: Verified whether `discount_applied` and `promo_code_used` were redundant and dropped `promo_code_used`.
- Database Integration: Connected the Python workflow to the database and loaded the cleaned DataFrame for SQL analysis.

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	location	size	color	season	review_rating	subscription_status	payment_method	shipping
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3900.000000	3900	3900	
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Credit Card	Free S
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	696	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.749949	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716223	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	

Figure 1. Python dataset preview and descriptive statistics.

subscription_status	payment_method	shipping_type	discount_applied	previous_purchases	preferred_payment_method	frequency_of_purchases	age_group	purchase_frequency_days
3900	3900	3900	3900	3900.000000		3900	3900	3353.000000
2	6	6	2	NaN		6	7	4
No	Credit Card	Free Shipping	No	NaN	PayPal	Every 3 Months	young_adult	NaN
2847	696	675	2223	NaN	677	584	1028	NaN
NaN	NaN	NaN	NaN	25.351538	NaN	NaN	NaN	101.390098
NaN	NaN	NaN	NaN	14.447125	NaN	NaN	NaN	124.140318
NaN	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	7.000000
NaN	NaN	NaN	NaN	13.000000	NaN	NaN	NaN	14.000000
NaN	NaN	NaN	NaN	25.000000	NaN	NaN	NaN	90.000000
NaN	NaN	NaN	NaN	38.000000	NaN	NaN	NaN	90.000000
NaN	NaN	NaN	NaN	50.000000	NaN	NaN	NaN	365.000000

Figure 2. Continued descriptive statistics and engineered fields.

4. Data Analysis using SQL

Structured SQL analysis was then performed to answer key business questions. The following sections present the question, the analytical approach, the result, and the supplied MySQL Workbench evidence.

4.1 Revenue by Gender

Business question: What is the total revenue generated by male versus female customers?

Male customers generated the larger total revenue in the supplied result set, while female customers contributed a smaller but still material share.

```

5      -- 1. What is the total revenue generated by male vs female customers?
6 •    select gender, SUM(purchase_amount_usd) as revenue
7      from customer
8      group by gender;
9
10     -- 2. Which customers used a discount but still spent more than the average purchase amount?
11 •    select customer_id, purchase_amount_usd

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

gender	revenue
Male	157890
Female	75191

Figure 3. SQL result — total revenue by gender.

4.2 High-Spending Discount Users

Business question: Which customers used a discount but still spent more than the average purchase amount?

The query identified 839 customers who met both conditions, showing that discount use was not limited to low-value transactions.

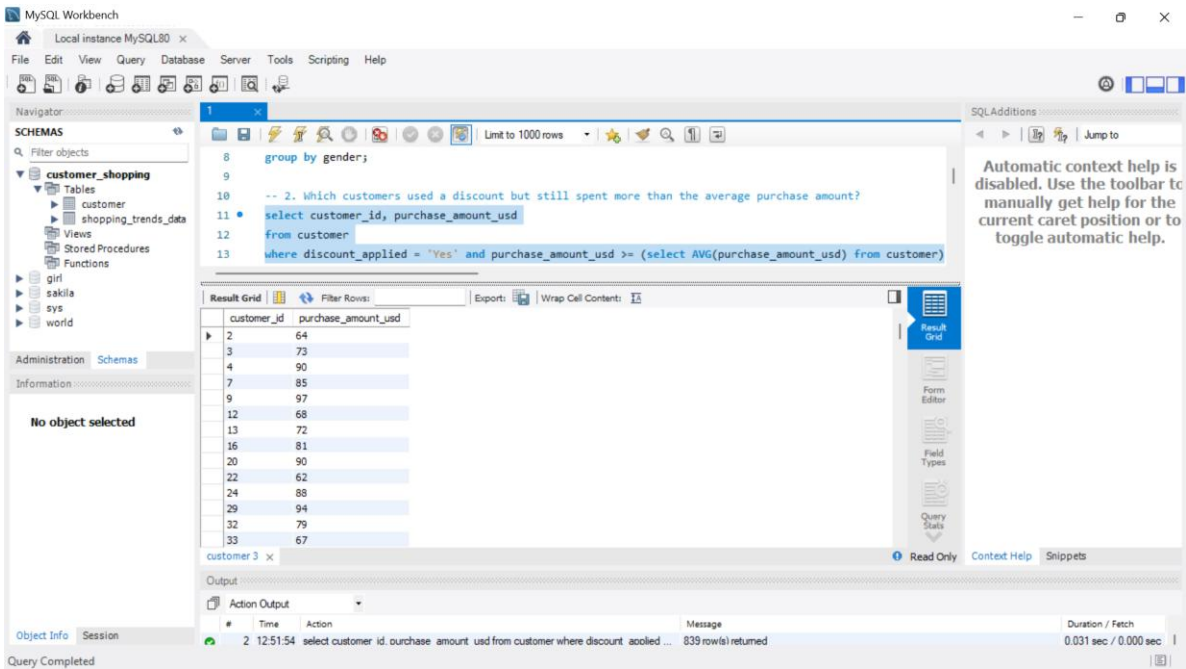


Figure 4. SQL result — discounted customers whose purchase exceeded the average.

4.3 Top 5 Products by Review Rating

Business question: Which five products have the highest average review rating?

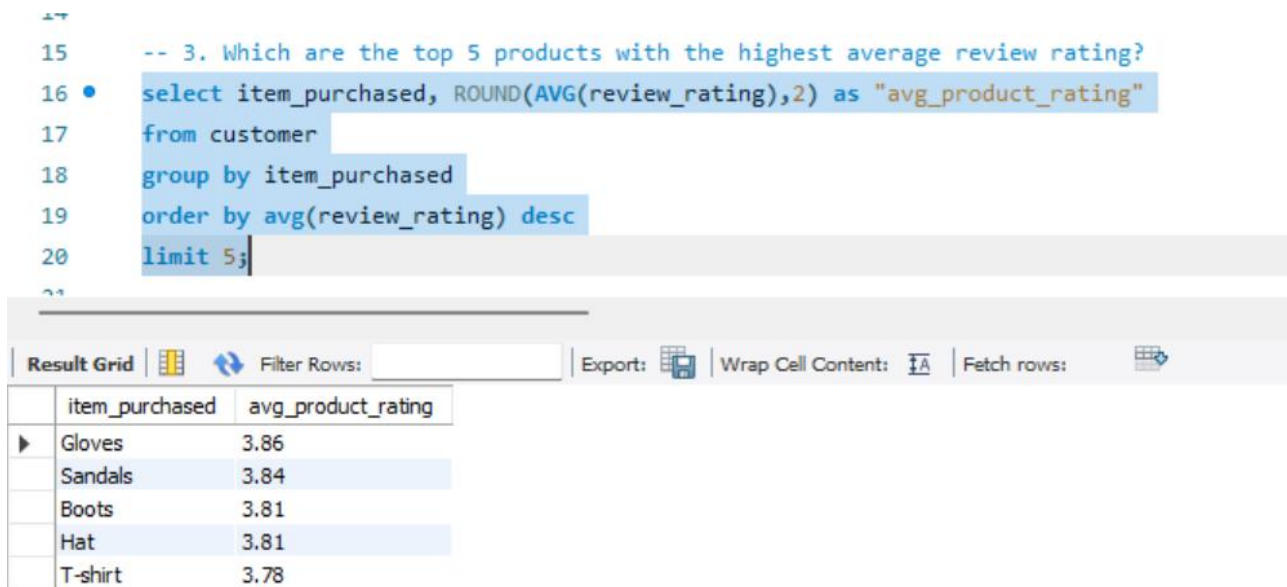


Figure 5. SQL query and result — top products by average review rating.

4.4 Average Purchase Amount by Shipping Type

Business question: How does the average purchase amount compare between Standard and Express shipping?

```

21
22  -- 4. Compare the average purchase amounts between standard and express shipping.
23 • select shipping_type,
24     ROUND(AVG(purchase_amount_usd),2)
25   from customer
26  where shipping_type in ('Standard', 'Express')
27  group by shipping_type;
28

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
shipping_type	ROUND(AVG(purchase_amount_usd),2)			
Express	60.48			
Standard	58.46			

Figure 6. SQL result — average purchase amount by shipping type.

The supplied results show that Express-shipping transactions have a higher average purchase amount.

4.5 Subscribers vs. Non-Subscribers

Business question: Do subscribed customers spend more? The analysis compares customer count, average spend and total revenue for subscribers and non-subscribers.

```

29  -- 5. Do subscribed customers spend more? Compare average spend and total revenue between
30  -- subscribers and non-subscribers.
31 • select subscription_status,
32     COUNT(customer_id) as total_customers,
33     ROUND(AVG(purchase_amount_usd),2) as avg_spend,
34     ROUND(SUM(purchase_amount_usd),2) as total_revenue
35   from customer
36  group by subscription_status
37  order by total_revenue, avg_spend desc;
38

```

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
subscription_status	total_customers	avg_spend	total_revenue				
Yes	1053	59.49	62645				
No	2847	59.87	170436				

Figure 7. SQL result — subscriber versus non-subscriber performance.

Subscribers have a slightly lower average spend in the supplied result, while non-subscribers generate much higher total revenue because they represent the larger customer group.

4.6 Discount-Dependent Products

Business question: Which five products have the highest percentage of purchases with discounts applied?

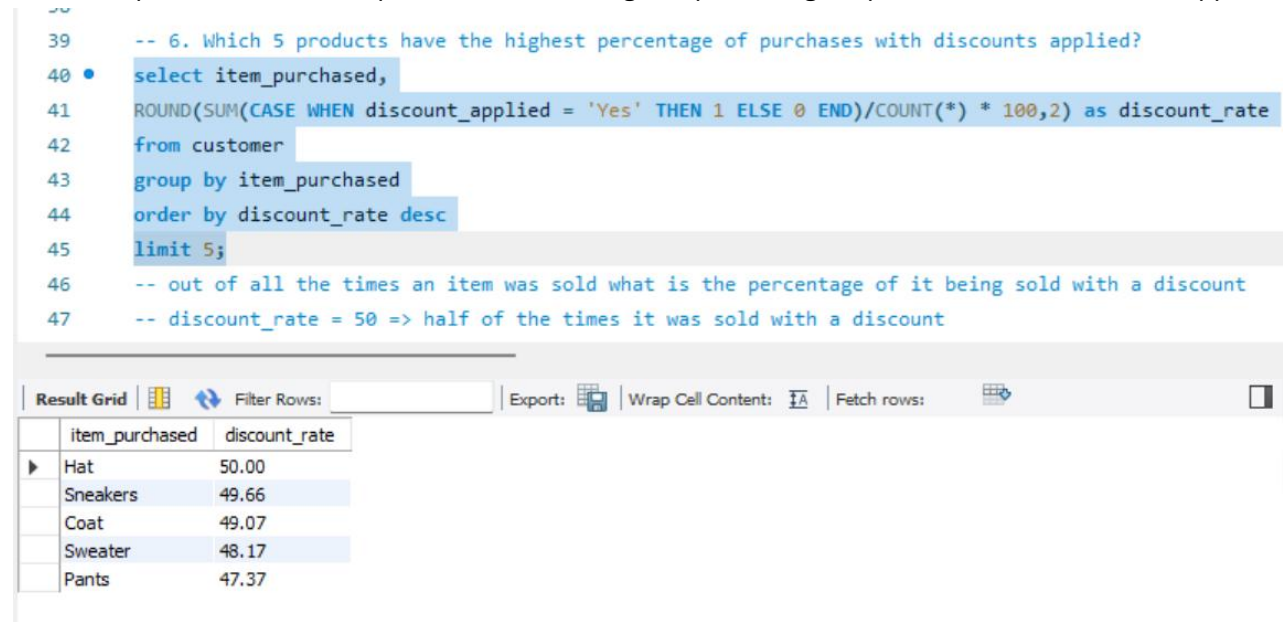


Figure 8. SQL result — products with the highest discount rates.

Hats have the highest discount incidence among the products shown, with half of the recorded purchases involving a discount.

4.7 Customer Segmentation

Customers were classified into New, Returning and Loyal segments using previous-purchase history.

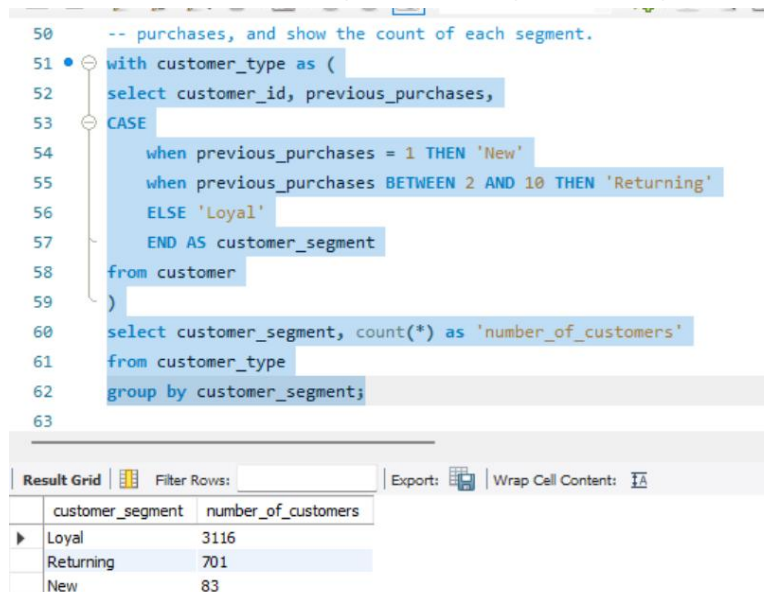


Figure 9. SQL result — customer segmentation by previous purchases

The dataset is dominated by the Loyal segment under the project's defined rules, followed by Returning customers and then New customers.

4.8 Top 3 Products per Category

Business question: Which products are the most purchased within each product category?

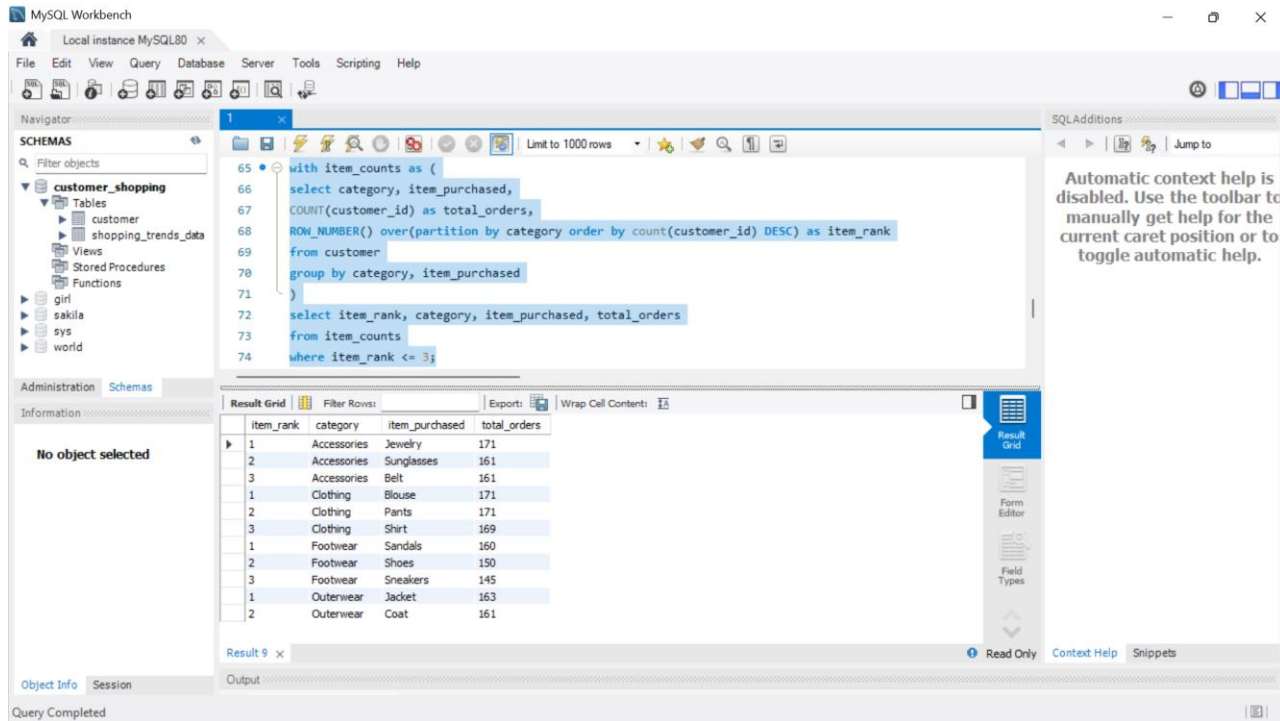


Figure 10. SQL result — top products by category.

4.9 Repeat Buyers and Subscription Status

Business question: Are customers with more than five purchases also likely to subscribe?

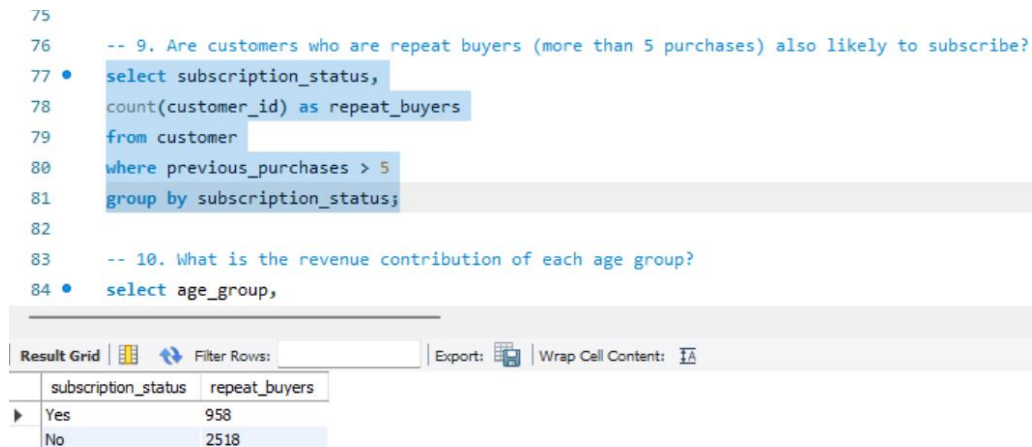


Figure 11. SQL result — repeat buyers according to subscription status.

Among customers with more than five previous purchases, the non-subscribed group is larger in absolute count than the subscribed group.

Figure 13. SQL result — repeat buyers by subscription status.

4.10 Revenue by Age Group

Business question: What is the revenue contribution of each age group?

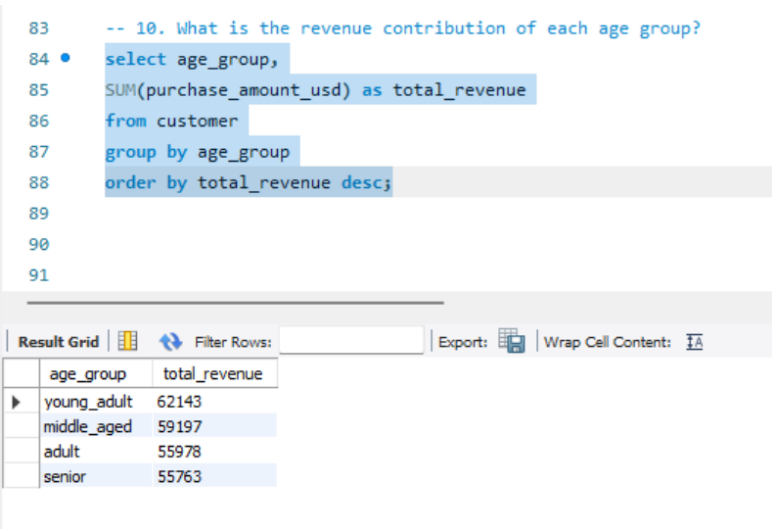


Figure 12. SQL result — revenue contribution by age group.

Young Adults contribute the highest revenue in the supplied results, followed by Middle-aged, Adult and Senior customers.

5. Power BI Dashboard

The final stage of the project was to build an interactive Power BI dashboard that brings the major findings together in a single visual interface. The supplied dashboard includes filters for subscription status, gender, category and shipping type, together with KPI cards and category/age-group charts.

Dashboard Element	Purpose
Number of Customers	Shows the overall customer count.
Average Purchase Amount	Summarizes the typical transaction value.
Average Review Rating	Provides a high-level view of product feedback.
Subscription Status	Shows the share of subscribers and non-subscribers.
Revenue by Category	Compares revenue contribution across product categories.
Sales by Category	Compares order volume across categories.
Revenue by Age Group	Highlights the age groups contributing the most revenue.
Sales by Age Group	Compares customer/order volume across age groups.



Figure 13. Power BI Customer Behaviour Dashboard.

6. Business Recommendations

- **Boost Subscriptions** — Promote exclusive benefits and clearer value propositions for subscribers.
- **Customer Loyalty Programs** — Reward repeat buyers and use targeted incentives to move customers into the Loyal segment.
- **Review Discount Policy** — Balance sales stimulation with margin control, particularly for products with high discount incidence.
- **Product Positioning** — Highlight top-rated and best-selling products in campaigns and merchandising.
- **Targeted Marketing** — Focus promotional efforts on high-revenue age groups and customers who prefer Express shipping.

These recommendations follow directly from the analysis presented in the supplied project reference: subscription participation is relatively low, repeat/loyal customers dominate the defined segments, and several products show strong ratings or high discount dependence.

7. Conclusion

The Customer Shopping Behaviour Analysis demonstrates an end-to-end analytics workflow: data preparation in Python, structured business analysis in SQL, and visual communication through Power BI. The analysis identifies meaningful differences in revenue by gender and age group, highlights high-rated products and frequently purchased products, compares shipping and subscription behaviour, and segments customers by purchasing history.

The resulting dashboard provides a concise way to monitor these patterns, while the SQL analysis supplies the underlying evidence needed for business recommendations. Together, the workflow provides a practical foundation for customer-focused marketing, loyalty, discount and product strategies.

Appendix A — Screenshot Reference

The following figures are the project screenshots supplied with the reference report. They are embedded throughout the report above at the relevant analytical steps.

Figure 1: Python dataset preview and descriptive statistics

Figure 2: Python descriptive statistics continuation

Figure 3: Revenue by gender

Figure 4: High-spending discount users

Figure 5: Top products by review rating — query and result

Figure 6: Top products by review rating — result evidence

Figure 7: Average purchase by shipping type

Figure 8: Subscribers versus non-subscribers

Figure 9: Products with highest discount rates

Figure 10: Customer segmentation

Figure 11: Top products per category

Figure 12: Revenue by age group / SQL continuation

Figure 13: Power BI dashboard